# Outdoor presence detection with radar sensor

Application note

## Versions

| Version | Date | Rationale |
|---|---|---|
| **1.0** | June 17, 2026 | First release. Author: ROJ, KOA |

## Legal disclaimer

The evaluation board is for testing purposes only and, because it has limited functions and limited resilience, is not suitable for permanent use under real conditions. If the evaluation board is nevertheless used under real conditions, this is done at one's responsibility; any liability of Rutronik is insofar excluded.

## Table of contents

# Introduction

This document describes the calculations behind the outside presence detection using Infineon BGT60TR13C radar sensor.

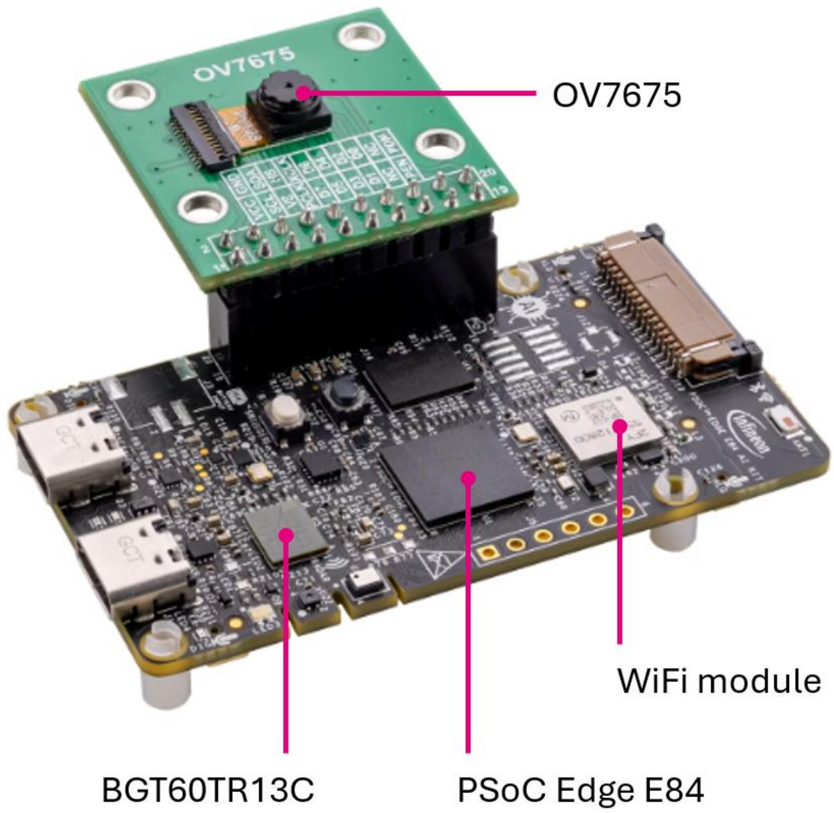
The challenge of this project lies in detecting the presence of a person in rainy conditions.

# Hardware and test conditions

## Infineon PSE84 AI kit

Infineon PSE84 AI kit was used for tests. It includes:

- OV7675 camera for visual control of the presence,
- BGT60TR13C radar sensor,
- Wifi module for data communication with computer.

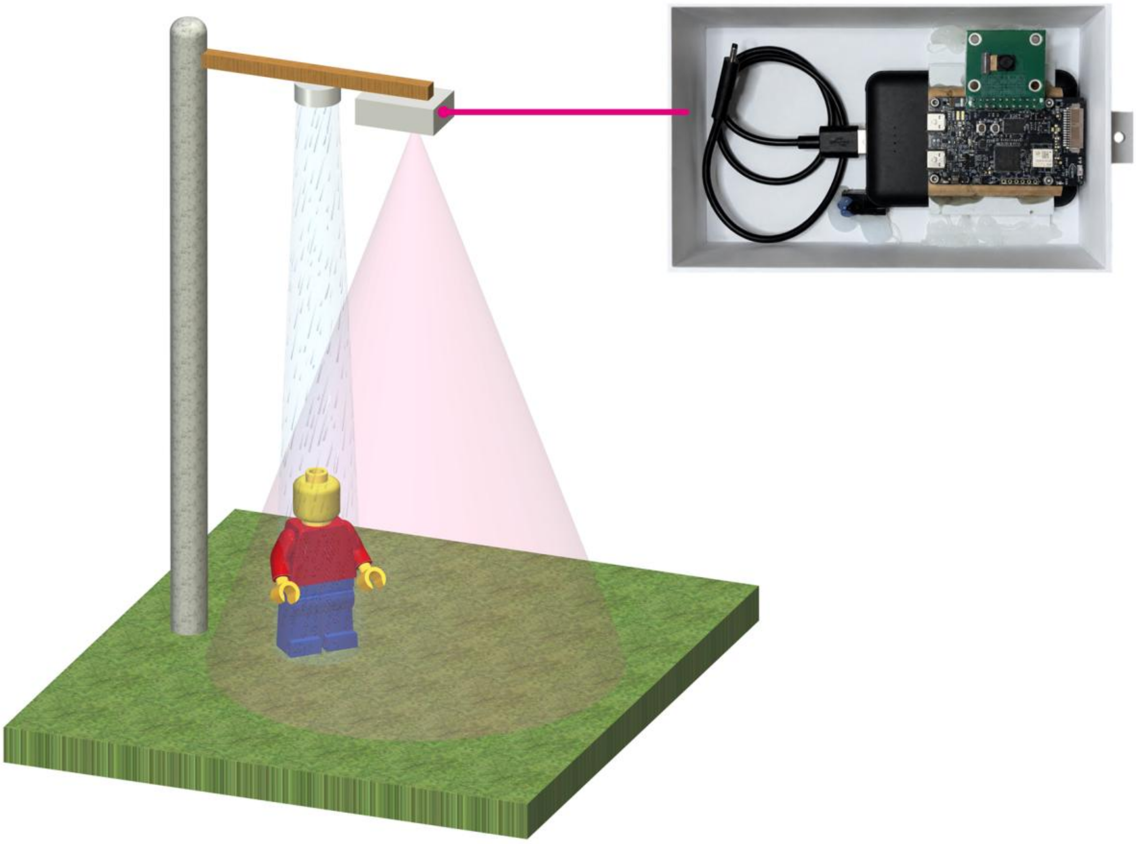


## Test setup

The Infineon PSE84 AI kit was placed inside a box with the radar pointing downward. The distance from the radar to the ground was 2.6 m.

A water source was set up next to the box containing the radar.

A person walked through the radar's field of view in various directions. The person also stood in the rain and walked through it.



# Software and firmware

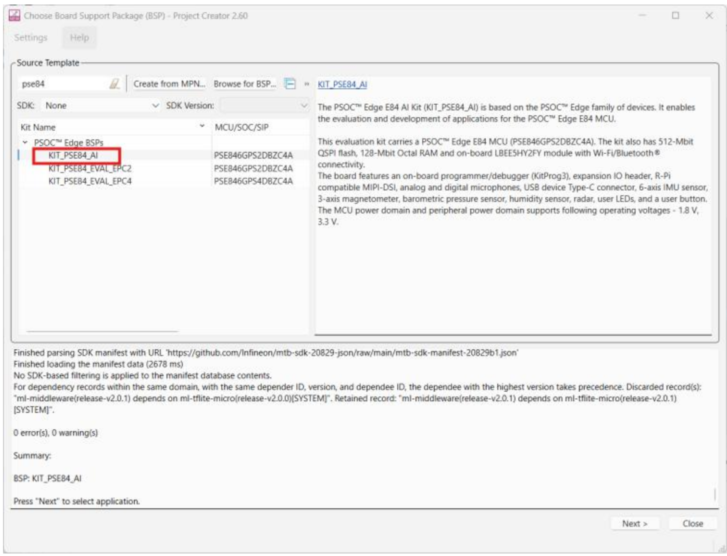
The firmware and GUI used for visualization and logging of the data is stored at our GitHub: https://github.com/RutronikSystemSolutions/kit_pse84_ai_sensor_wifi_streaming

## Firmware

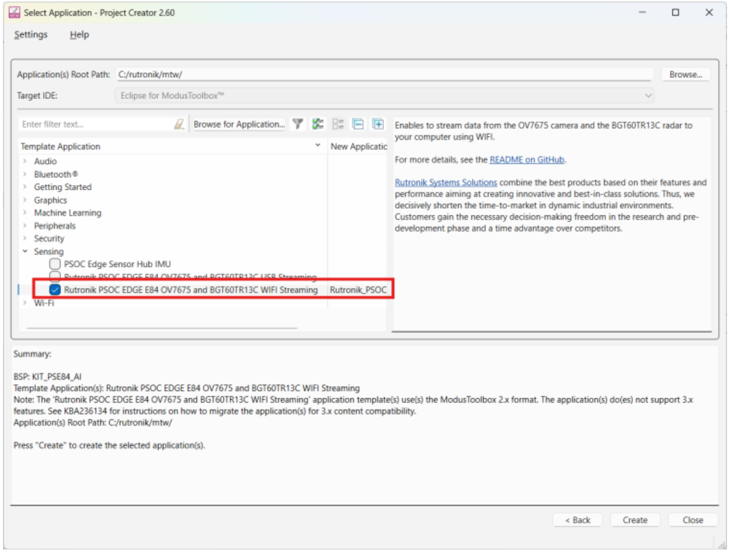
### Modus Toolbox integration

The firmware example is available directly inside Modus Toolbox.

In project creator,
select **KIT_PSE84_AI**.




In the **Sensing** group,
select the marked project.




### Wifi connection mode

The device communicates over Wifi with the computer. You can use the device in 2 modes: access point mode or not.

The configuration is done in the proj_cm55/tcp_server/network_credentials.h file.

### Radar settings

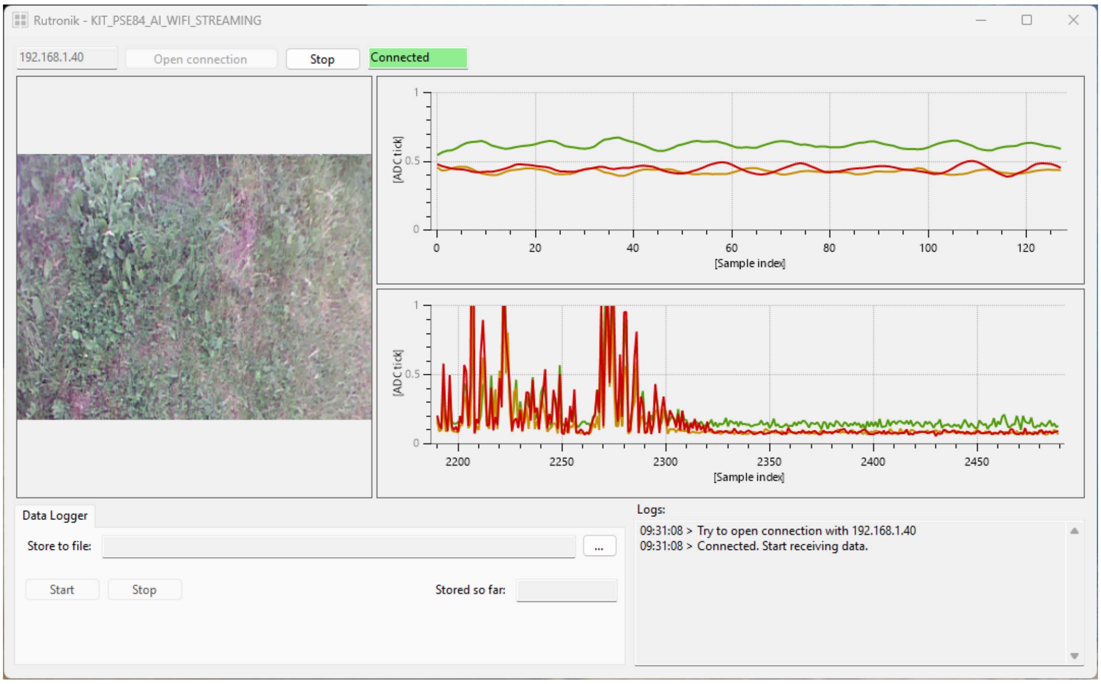
At the start, the BGT60TR13C is initialized and starts streaming data.

The configuration of the radar is stored inside the file proj_cm55/driver/radar_settings.h. This file has been generated using the “Radar Fusion GUI” software from Infineon.

You can change the configuration of the radar, but if you do it, please also change the configuration in the GUI.

## GUI

### Overview




Enter the IP address of the device, and then click on the **Open connection** button. Data will be streamed automatically and displayed.

The picture is received from the OV7675 camera.

### Radar configuration

If you change the radar configuration in the firmware, then you will also have to change it in the GUI.

This is done in the file ProjectConfiguration.cs.

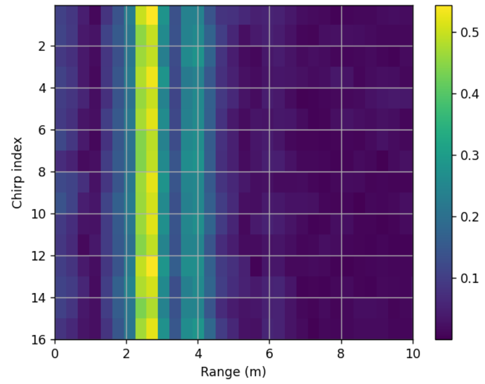
```
public const int RADAR_ANTENNA_COUNT = 3;
public const int RADAR_SAMPLES_PER_CHIRP = 128;
public const int RADAR_CHIRPS_PER_FRAME = 16;
public const int RADAR_BYTES_PER_SAMPLE = 2;
```

# Presence detection and water effect

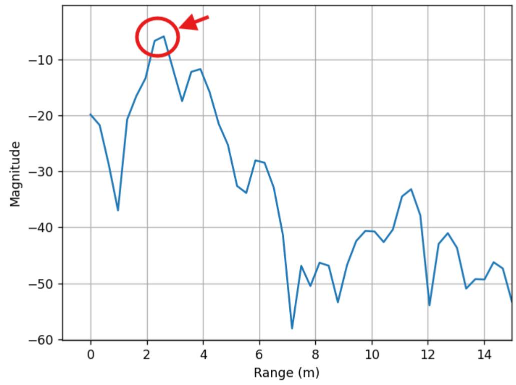
We followed the next scenario:

- No water
- Start spraying water
- Someone enters radar’s field of view
- Someone leaves radar’s field of view
- Stop spraying water
- No water
-

When no water and no presence, the radar data shows a static target at 2,6 m distance (ground level).



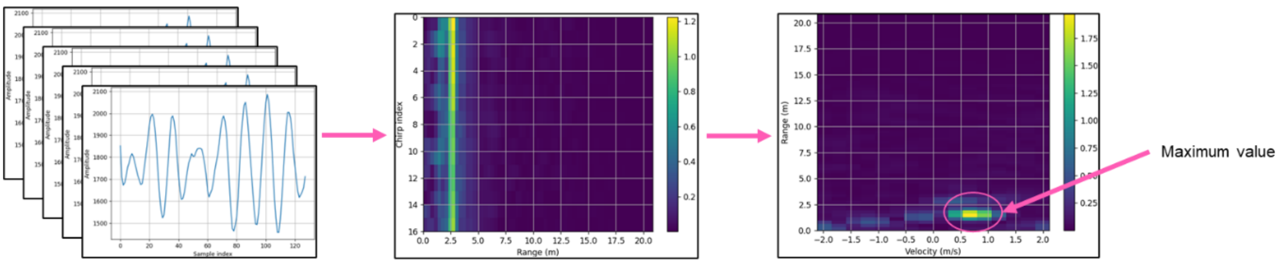

Magnitude of the range FFT for all the chirps of the first frame.
Yellow color (over all the chirps) indicates a static target.




Magnitude of the averaged range FFT for the first frame.
Reflection of ground is marked in red.

## Presence detection

For presence detection, we displayed the magnitude of the biggest motion (extracted from Doppler FFT) for each frame. For each frame we extract only one value: the “maximum value” as explained in the picture below.

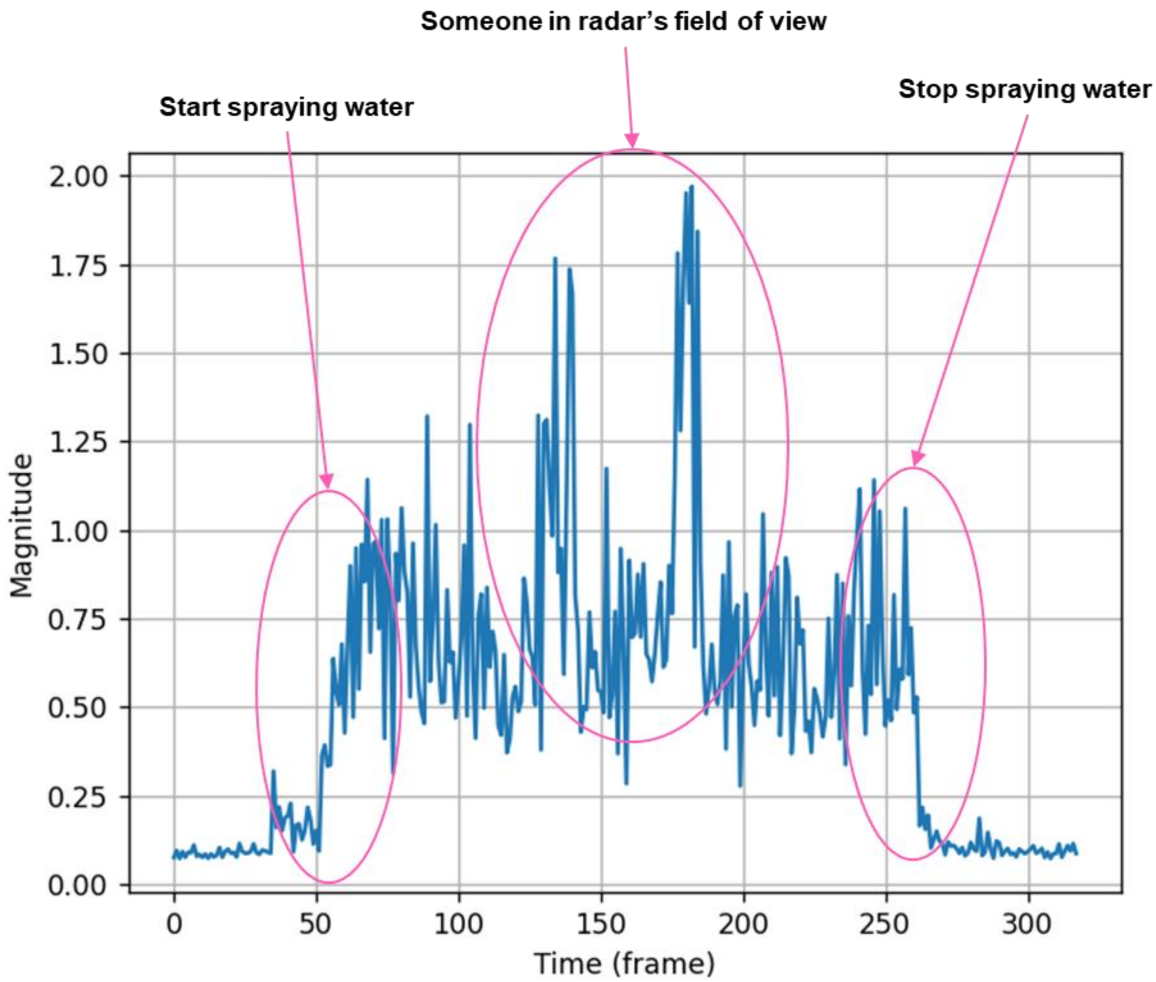



Time signal (frame)
N chirps of X samples

Range FFT

Doppler FFT

Here the maximum value per frame is displayed when the scenario is completed.

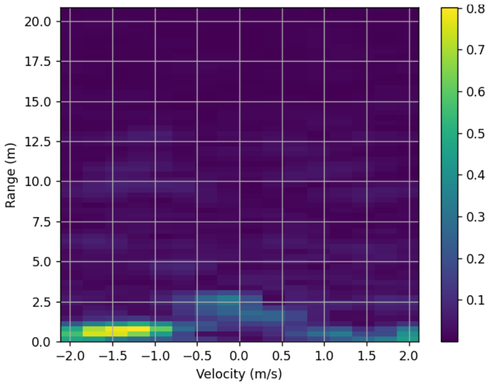



We observe 2 effects:

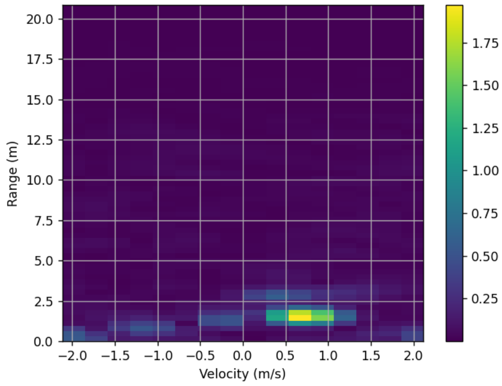
- Water spray is detected by the radar sensor (average magnitude of around 0.75).
- Presence is also detected, and maximum amplitude of the detected presence is around 0.75.

Thus, the presence effect is only 2 times bigger than the water effect.

## Discriminate between water and presence

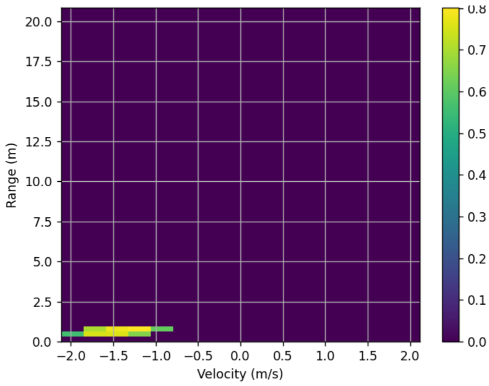


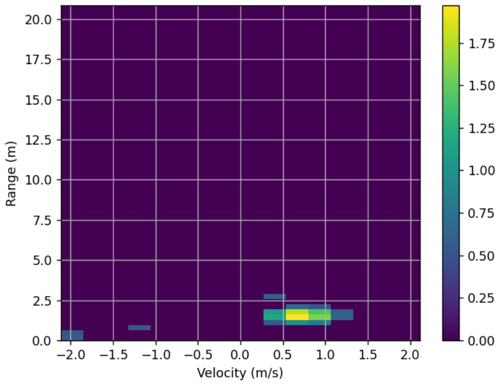
Effect of water/rain:
produces signal with high velocities



Effect of someone in radar's field of view:
velocity smaller that water/rain

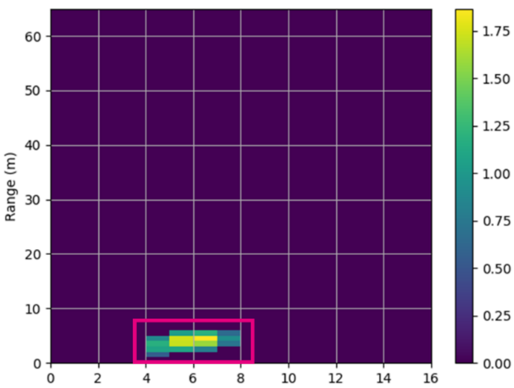
If we remove the signal below the threshold (0.5), we also observe that the areas for rain are smaller than the areas for a person (if looking at the area covered for range).



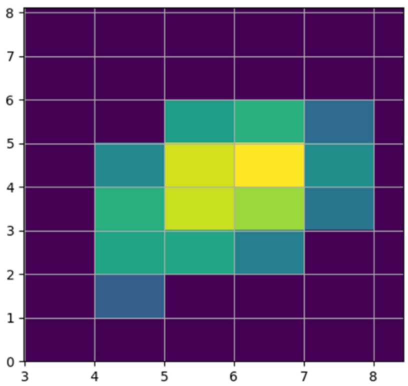




The algorithm to discriminate water/rain from person presence will be as follow:

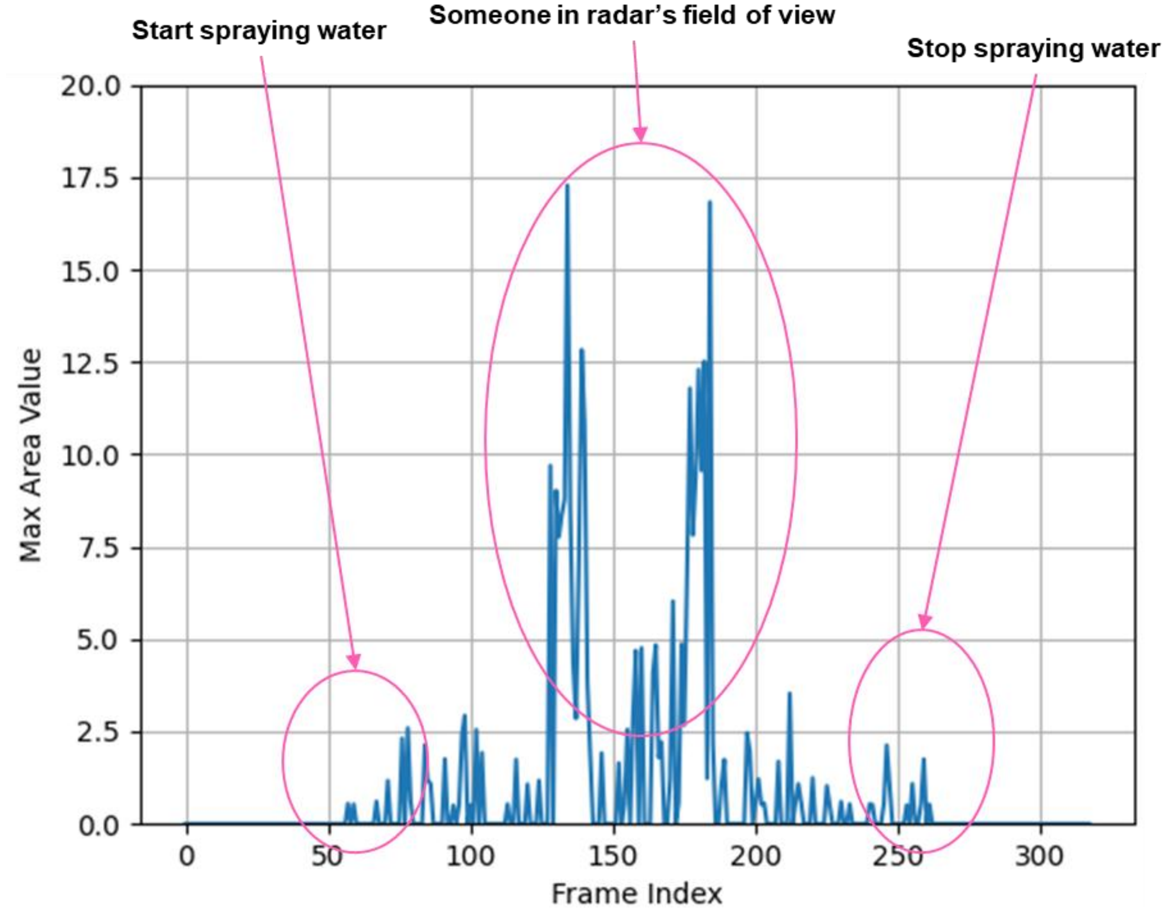
- Remove high velocities motion in Doppler FFT.
- Remove everything below threshold in Doppler FFT.
- Instead of just extracting the maximum value, extract the sum of the values inside the biggest area.



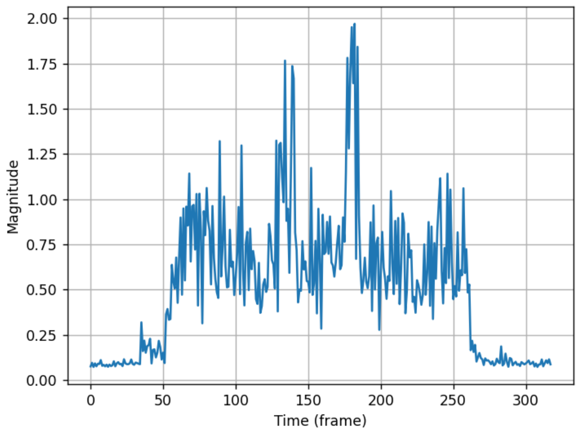
Doppler FFT after removing high velocities
(unit of X axis is the index
in the Doppler FFT matrix)


Zoom on the biggest area.
The value that is extracted per frame is the
sum of all the values of the zoomed area.

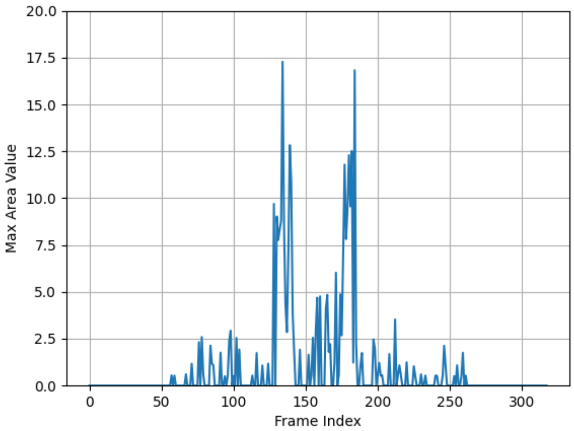
Applying this new algorithm to our previous scenario results in such a graph.




The presence now becomes obvious.




Standard maximum value algorithm




New algorithm enabling
rain/water and person presence discrimination.